

[Vitamin E: Fact Sheet for Consumers \(original version\)](#)

Vitamin E: Fact Sheet for Consumers

Table of Contents

- [What is vitamin E and what does it do?](#)
- [How much vitamin E do I need?](#)
- [What foods provide vitamin E?](#)
- [What kinds of vitamin E dietary supplements are available?](#)
- [Am I getting enough vitamin E?](#)
- [What happens if I don't get enough vitamin E?](#)
- [What are some effects of vitamin E on health?](#)
- [Can vitamin E be harmful?](#)
- [Does vitamin E interact with medications or other dietary supplements?](#)
- [Vitamin E and healthful eating](#)
- [Where can I find out more about vitamin E?](#)
- [Disclaimer](#)
- [Glossary](#)

What is vitamin E and what does it do?

Vitamin E is a fat-soluble nutrient found in many foods. In the body, it acts as an antioxidant, helping to protect cells from the damage caused by free radicals. Free radicals are compounds formed when our bodies convert the food we eat into energy. People are also exposed to free radicals in the environment from cigarette smoke, air pollution, and ultraviolet light from the sun.

The body also needs vitamin E to boost its immune system so that it can fight off invading bacteria and viruses. It helps to widen blood vessels and keep blood from clotting within them. In addition, cells use vitamin E to interact with each other and to carry out many important functions.

How much vitamin E do I need?

The amount of vitamin E you need each day depends on your age. Average daily recommended amounts are listed below in milligrams (mg).

Life Stage	Recommended Amount
Birth to 6 months	4 mg
Infants 7–12 months	5 mg
Children 1–3 years	6 mg
Children 4–8 years	7 mg
Children 9–13 years	11 mg
Teens 14–18 years	15 mg
Adults	15 mg
Pregnant teens and women	15 mg
Breastfeeding teens and women	19 mg

What foods provide vitamin E?

Vitamin E is found naturally in foods and is added to some fortified foods. You can get recommended amounts of vitamin E by eating a variety of foods including the following:

- Vegetable oils like wheat germ, sunflower, and safflower oils are among the best sources of vitamin E. Corn and soybean oils also provide some vitamin E.
- Nuts (such as peanuts, hazelnuts, and, especially, almonds) and seeds (like sunflower seeds) are also among the best sources of vitamin E.
- Green vegetables, such as spinach and broccoli, provide some vitamin E.
- Food companies add vitamin E to some breakfast cereals, fruit juices, margarines and spreads, and other foods. To find out which ones have vitamin E, check the product labels.

What kinds of vitamin E dietary supplements are available?

Vitamin E supplements come in different amounts and forms. Two main things to consider when choosing a vitamin E supplement are:

1. The amount of vitamin E: Most once-daily multivitamin/mineral supplements provide about 13.5 mg of vitamin E, whereas vitamin E-only supplements commonly contain 67 mg or more. The doses in most vitamin E-only supplements are much higher than the recommended amounts. Some people take large doses because they believe or hope that doing so will keep them healthy or lower their risk of certain diseases.
2. The form of vitamin E: Although vitamin E sounds like a single substance, it is actually the name of eight related compounds in food, including alpha-tocopherol. Each form has a different potency, or level of activity in the body.

Vitamin E from natural sources is commonly listed as *d*-alpha-tocopherol on food packaging and supplement labels. Synthetic (laboratory-made) vitamin E is commonly listed as *dl*-alpha-tocopherol. The natural form is more potent; 1 mg vitamin E = 1 mg *d*-alpha-tocopherol (natural vitamin E) = 2 mg *dl*-alpha-tocopherol (synthetic vitamin E).

Some food and dietary supplement labels still list vitamin E in International Units (IUs) rather than mg. One IU of the natural form of vitamin E is equivalent to 0.67 mg. One IU of the synthetic form of vitamin E is equivalent to 0.45 mg.

Some vitamin E supplements provide other forms of the vitamin, such as gamma-tocopherol, tocotrienols, and mixed tocopherols. Scientists do not know if any of these forms are superior to alpha-tocopherol in supplements.

Am I getting enough vitamin E?

The diets of most Americans provide less than the recommended amounts of vitamin E. Nevertheless, healthy people rarely show any clear signs that they are not getting enough vitamin E (see next question for information on the signs of vitamin E deficiency).

What happens if I don't get enough vitamin E?

Vitamin E deficiency is very rare in healthy people. It is almost always linked to certain diseases in which fat is not properly digested or absorbed. Examples include Crohn's disease, cystic fibrosis, and certain rare genetic diseases such as abetalipoproteinemia and ataxia with vitamin E deficiency (AVED). Vitamin E needs some fat for the digestive system to absorb it.

Vitamin E deficiency can cause nerve and muscle damage that results in loss of feeling in the arms and legs, loss of body movement control, muscle weakness, and vision problems. Another sign of deficiency is a weakened immune system.

What are some effects of vitamin E on health?

Scientists are studying vitamin E to understand how it affects health. Here are several examples of what this research has shown.

Heart disease

Some studies link higher intakes of vitamin E from supplements to lower chances of developing coronary heart disease. However, the best research finds no benefit. People in these studies are randomly assigned to take vitamin E or a placebo (dummy pill with no vitamin E or active ingredients) and they don't know which they are taking. Vitamin E supplements do not seem to prevent heart disease, reduce its severity, or affect the risk of death from this disease. Scientists do not know whether high intakes of vitamin E might protect the heart in younger, healthier people who do not have a high risk of heart disease.

Cancer

Most research indicates that vitamin E does not help prevent cancer and may be harmful in some cases. Large doses of vitamin E have not consistently reduced the risk of colon and breast cancer in studies, for example. A large study found that taking vitamin E supplements (180 mg/day [400 IU]) for several years increased the risk of developing prostate cancer in men. Two studies that followed middle-age men and women for 7 or more years found that extra vitamin E (201–268 mg/day [300–400 IU], on average) did not protect them from any form of cancer. However, one study found a link between the use of vitamin E supplements for 10 years or more and a lower risk of death from bladder cancer.

Vitamin E dietary supplements and other antioxidants might interact with chemotherapy and radiation therapy. People undergoing these treatments should talk with their doctor or oncologist before taking vitamin E or other antioxidant supplements, especially in high doses.

Eye disorders

Age-related macular degeneration (AMD), or the loss of central vision in older people, and cataracts are among the most common causes of vision loss in older people. The results of research on whether vitamin E can help prevent these conditions are inconsistent. Among people with AMD who were at high risk of developing advanced AMD, a supplement containing large doses of vitamin E combined with other antioxidants, zinc, and copper showed promise for slowing down the rate of vision loss.

Mental function

Several studies have investigated whether vitamin E supplements might help older adults remain mentally alert and active as well as prevent or slow the decline of mental function and Alzheimer's disease. So far, the research provides little evidence that taking vitamin E supplements can help healthy people or people with mild mental functioning problems to maintain brain health.

Can vitamin E be harmful?

Vitamin E that is naturally present in food and beverages is not harmful and does not need to be limited.

In supplement form, however, high doses of vitamin E might increase the risk of bleeding (by reducing the blood's ability to form clots after a cut or injury) and of serious bleeding in the brain (known as hemorrhagic stroke). Because of this risk, the upper limit for adults is 1,000 mg/day for supplements of either natural or synthetic vitamin E. This is equal to 1,500 IU/day for natural vitamin E supplements and 1,100 IU/day for synthetic vitamin E supplements. The upper limits for children are lower than those for adults. Some research suggests that taking vitamin E supplements even below these upper limits might cause harm. In one study, for example, men who took 400 IU (180 mg) of synthetic vitamin E each day for several years had an increased risk of prostate cancer.

Does vitamin E interact with medications or other dietary supplements?

Vitamin E dietary supplements can interact or interfere with certain medicines that you take. Here are some examples:

- Vitamin E can increase the risk of bleeding in people taking anticoagulant or antiplatelet medicines, such as warfarin (Coumadin).
- In one study, vitamin E plus other antioxidants (such as vitamin C, selenium, and beta-carotene) reduced the heart-protective effects of two drugs taken in combination (a statin and niacin) to affect blood cholesterol levels.
- Taking antioxidant supplements while undergoing chemotherapy or radiation therapy for cancer could alter the effectiveness of these treatments.

Tell your doctor, pharmacist, and other health care providers about any dietary supplements and medicines you take. They can tell you if those dietary supplements might interact or interfere with your prescription or over-the-counter medicines, or if the medicines might interfere with how your body absorbs, uses, or breaks down nutrients.

Vitamin E and healthful eating

People should get most of their nutrients from food and beverages, according to the federal government's *Dietary Guidelines for Americans*. Foods contain vitamins, minerals, dietary fiber and other components that benefit health. In some cases, fortified foods and dietary supplements are useful when it is not possible to meet needs for one or more nutrients (for example, during specific life stages such as pregnancy). For more information about building a healthy dietary pattern, see the [Dietary Guidelines for Americans](#) and the U.S. Department of Agriculture's (USDA's) [MyPlate](#).

Where can I find out more about vitamin E?

- For general information on vitamin E
 - Office of Dietary Supplements (ODS) Health Professional Fact Sheet on [Vitamin E](#)
 - [Vitamin E](#), MedlinePlus
- For more information on food sources of vitamin E
 - USDA's [FoodData Central](#)
 - Nutrient List for vitamin E (listed by [food](#) or by [vitamin E content](#)), USDA
- For more advice on choosing dietary supplements
 - ODS [Frequently Asked Questions: Which brand\(s\) of dietary supplements should I purchase?](#)
- For information about building a healthy dietary pattern
 - [MyPlate](#)
 - [Dietary Guidelines for Americans](#)

Glossary

abetalipoproteinemia

A rare inherited disease in which the intestine cannot absorb fats and fat-soluble vitamins from food. It is associated with fatty stools, diarrhea, nerve problems, and eye disease.

absorption

In nutrition, the process of moving protein, carbohydrates, fats, and other nutrients from the digestive system into the bloodstream. Most absorption occurs in the small intestine.

age-related macular degeneration

AMD. An eye disease that results in a loss of central, "straight-ahead" vision. AMD is the leading cause of vision loss in older Americans.

alpha-tocopherol

The form of vitamin E that is found in the largest amount in humans and is the most active form of vitamin E. It is an antioxidant.

Alzheimer's disease

A brain disease in which thinking, memory, and reasoning ability is slowly destroyed. In advanced stages, an affected person becomes disoriented and confused, has mood and behavior changes, and has difficulty talking, walking, and swallowing. Alzheimer's disease is progressive, irreversible, and incurable.

anticoagulant

A drug or other substance that stops blood clots from forming. Also called a blood thinner.

antioxidant

A substance that protects cells from damage caused by free radicals (compounds formed during the metabolism of oxygen). It may help prevent the development of some chronic diseases such as cancer.

Antioxidants include beta-carotene; lutein; lycopene; vitamins A, C, and E; selenium; and zinc.

ataxia

Loss of muscle coordination.

bacteria

Single-celled organisms that are too small to be seen without a microscope. Bacteria are found everywhere and may be helpful or harmful.

beta-carotene

A carotenoid found in carrots, cantaloupe, apricots, sweet potatoes, pumpkin, winter squash, mangos, collard greens, spinach, kale, broccoli, and other orange, red, and dark green fruits and vegetables.

bladder cancer

Uncontrolled growth of abnormal cells in the bladder (the organ that stores urine).

blood vessel

A tube through which blood circulates in the body. Blood vessels include a network of arteries, arterioles, capillaries, venules, and veins.

breast cancer

Uncontrolled growth of abnormal cells in the breast.

cancer

A group of diseases in which cells divide abnormally and without control, and spread to nearby tissues and other parts of the body. Without treatment, cancer can stop organs from working normally, damage body systems, and cause the patient to die. Cancer may be caused by multiple factors, such as radiation, sunlight, tobacco, certain viruses, and poisonous chemicals; however, the cause of many cancers is unknown.

cataract

A condition in which the lens of the eye becomes cloudy. Symptoms include blurred, cloudy, or double vision; sensitivity to light; and difficulty seeing at night. Without treatment, cataracts can cause blindness. Cataracts occur in people of all ages but are most common in the elderly.

cell

The individual unit that makes up the tissues of the body. All living things are made up of one or more cells, which are the smallest units of living structure capable of independent existence.

chemotherapy

A chemical that kills bacteria, viruses, fungi, or tumor cells. It usually refers to drugs used in cancer treatment.

cholesterol

A substance found throughout the body. It is made by the liver and is an important component of cells. Cholesterol is also used to make hormones, bile acid, and vitamin D. Foods that come from animals contain cholesterol, including eggs, dairy products, meat, poultry and fish. High blood levels of cholesterol increase a person's chance (risk) of developing atherosclerosis and heart disease.

colon

A tube-like organ about 5 feet long in adults that is connected to the small intestine at one end and the anus at the other. The colon absorbs water, some nutrients, and electrolytes (such as sodium and chloride) from partially digested food. The remaining material (solid waste called stool) moves through the colon to the rectum and leaves the body through the anus as a bowel movement. The colon is part of the digestive system (a series of organs from the mouth to the anus). Also called the large intestine.

compound

In pharmacy, a substance that contains more than one ingredient.

control

In a clinical trial, the group of participants that does not receive the new treatment being studied. This group is compared with the group receiving the new treatment, to see whether the new treatment works. In an observational study, the controls are participants who do not have a particular health condition; the control group is compared with the group of participants who do have the condition to see if certain factors (such as diet, activity level, or use of dietary supplements) may be associated with developing or preventing the condition.

copper

In nutrition, a mineral the body needs (along with iron) to make red blood cells. Copper also helps keep the immune system, blood vessels, nerves, and bones healthy. Copper is found in some foods, including oysters and other shellfish, whole grains, beans, nuts, potatoes, organ meats, dark leafy greens, and dried fruits.

coronary heart disease

A disease in which the blood vessels (coronary arteries) that carry blood and oxygen to the heart are narrowed or blocked, which can cause chest pain, shortness of breath, and heart attack. It is usually caused by a build-up of fat and cholesterol deposits inside the arteries (atherosclerosis).

Crohn's disease

A long-lasting (chronic) disease that causes severe irritation in the gastrointestinal tract. It usually affects the lower small intestine (called the ileum) or the colon, but it can affect any part of the digestive tract from the mouth to the anus. It is painful, causing severe watery or bloody diarrhea, and may lead to life-threatening complications. Crohn's disease is a form of inflammatory bowel disease.

cystic fibrosis

A common inherited disease that causes the body to make thick, sticky mucus that builds up in the lungs and blocks the airways, leading to repeated serious lung infections. Mucus also blocks the pancreas, which stops digestive enzymes from reaching the intestines. Cystic fibrosis also causes very salty sweat, which can lead to dehydration, increased heart rate, tiredness, low blood pressure, and heat stroke.

deficiency

An amount that is not enough; a shortage.

dietary fiber

A substance in plants that you cannot digest. It adds bulk to your diet to make you feel full, helps prevent constipation, and may help lower the risk of heart disease and diabetes. Good sources of dietary fiber include whole grains (such as brown rice, oats, quinoa, bulgur, and popcorn), legumes (such as black beans, garbanzo beans, split peas, and lentils), nuts, seeds, fruit, and vegetables.

Dietary Guidelines for Americans

Advice from the federal government to promote health and reduce the chance (risk) of long-lasting (chronic) diseases through nutrition and physical activity. The Guidelines are updated and published every 5 years by the US Department of Health and Human Services and the US Department of Agriculture.

dietary supplement

A product that is intended to supplement the diet. A dietary supplement contains one or more dietary ingredients (including vitamins, minerals, herbs or other botanicals, amino acids, and other substances) or their components; is intended to be taken by mouth as a pill, capsule, tablet, or liquid; and is identified on the front label of the product as being a dietary supplement.

dose

The amount of medicine or other substance taken at one time or over a specific period of time.

drug

Any substance (other than food) that is used to prevent, diagnose, treat, or relieve symptoms of a disease or abnormal condition. Also, a substance that alters mood or body function or that can be habit-forming or addictive, especially a narcotic.

effectiveness

In medicine, the ability of an intervention (for example, a drug, surgery, or a dietary supplement) to produce the desired beneficial effect under the usual conditions of care by a health care provider.

evidence

Information used to support the use of a particular screening procedure, treatment, or preventive measure. In medicine, evidence needed to determine effectiveness is provided by laboratory research, clinical trials, and other studies.

fat soluble

Able to be dissolved in fat.

fortified

When nutrients (such as vitamins and minerals) are added to a food product. For example, when calcium is added to orange juice, the orange juice is said to be "fortified with calcium". Similarly, many breakfast cereals are "fortified" with several vitamins and minerals.

free radical

An atom or molecule made in the body that can damage cells. A free radical has at least one unpaired electron, which makes it unstable. To become stable, the free radical takes an electron away from another atom, which makes *that* atom unstable, and starts a chain reaction that can injure cells. Free radicals are made during chemical changes that take place in a cell or an organism to produce energy and basic materials needed for important life processes (metabolism). They also come from tobacco smoke, pollution, radiation from the sun and x-rays, and other sources outside the body. Free radicals damage cells, cause genetic alterations (mutations), and may play a role in cancer, heart disease, and age-related diseases (such as Alzheimer's, Parkinson's, and Lou Gehrig's diseases). Free radicals are also beneficial; they are involved in killing germs (microorganisms) and they help hormones and chemical messengers communicate with cells. Proteins (enzymes) made by the body, and vitamin C, vitamin E, and beta carotene in the diet help prevent free radical damage.

gene

The functional and physical unit of heredity passed from parent to offspring. Genes are pieces of DNA, and most genes contain the information for making a specific protein.

immune system

A group of organs and cells that defends the body against infection, disease, and altered (mutated) cells. It includes the thymus, spleen, lymphatic system (lymph nodes and lymph vessels), bone marrow, tonsils, and white blood cells.

infant

A child younger than 12 months old.

ingredient

In a dietary supplement, an ingredient is a component of the product, such as the main nutrient (vitamin, mineral, herb, amino acid, or enzyme) or any binder, color, filler flavor, or sweetener. In herbal supplements, the common name and Latin name (the genus and species) of the plant is given in the ingredient list. On a dietary supplement label, the ingredients are listed by weight, with the ingredient used in the largest amount first on the list and the ingredient used in the least amount at the end of the list.

interaction

A change in the way a dietary supplement acts in the body when taken with certain other supplements, medicines, or foods, or when taken with certain medical conditions. Interactions may cause the dietary supplement to be more or less effective, or cause effects on the body that are not expected.

International Unit

IU. A measurement used to measure the activity of some vitamins and other biological substances (such as enzymes and hormones).

IU

International Unit. A measurement used to measure the activity of some vitamins and other biological substances (such as enzymes and hormones).

label

When referring to dietary supplements, information that appears on the product container, including a descriptive name of the product stating that it is a "supplement"; the name and place of business of the manufacturer, packer, or distributor; a complete list of ingredients; and each dietary ingredient contained in the product. Supplements must also include directions for use, nutrition labeling in the form of a Supplement Facts panel that identifies each dietary ingredient contained in the product and the serving size, amount, and active ingredients.

milligram

mg. A measure of weight. It is a metric unit of mass equal to 0.001 gram (it weighs 28,000 times less than an ounce).

mineral

In nutrition, an inorganic substance found in the earth that is required to maintain health.

neuron

A nerve cell. Neurons send chemical and electrical messages throughout the nervous system that direct the body to function, move, think, and have emotions.

niacin

A nutrient that is needed by the body to make energy from food, and for important cellular functions. Niacin is found in some foods, including poultry, beef, pork, fish, nuts, legumes, whole grains, and enriched or fortified breads and cereals. In medicine it is used as a drug to lower high blood cholesterol levels. It is one of the B vitamins. Also called nicotinamide and nicotinic acid.

nutrient

A chemical compound in food that is used by the body to function and maintain health. Examples of nutrients include proteins, fats, carbohydrates, vitamins, and minerals.

Office of Dietary Supplements

ODS, Office of Disease Prevention, Office of Director, National Institutes of Health, Department of Health and Human Services. ODS strengthens knowledge and understanding of dietary supplements by evaluating scientific information, stimulating and supporting research, disseminating research results, and educating the public to foster an enhanced quality of life and health for the US population.

pharmacist

A person licensed to make and dispense (give out) prescription drugs and who has been taught how they work, how to use them, and their side effects.

placebo

An inactive substance or treatment that has no effect on the body and that ideally looks, smells, and tastes the same as, and is given the same way as, the active drug or treatment being tested. The effects of the active substance or treatment are compared to the effects of the placebo.

prescription

A written order from a health care provider for medicine, therapy, or tests.

prevent

To stop from happening.

prostate cancer

Uncontrolled growth of abnormal cells in the prostate (a gland in the male reproductive system found below the bladder and in front of the rectum). Prostate cancer usually occurs in older men.

radiation therapy

The use of high-energy radiation from x-rays and other sources to kill cancer cells, shrink tumors, and treat other conditions.

risk

The chance or probability that a harmful event will occur. In health, for example, the chance that someone will develop a disease or condition.

selenium

A mineral required in very small amounts to make important enzymes that are essential for good health. Selenium is found in some foods, including plant foods grown in selenium-rich soil, and some meats and seafood.

sign

An indication of disease that can be seen and/or measured. Examples include high fever, high blood pressure, infection, and coughing up blood.

soy

A plant that produces beans used in many food products. Soy products contain isoflavones (estrogen-like substances) that are being studied in the prevention of cancer, hot flashes that occur with menopause, and osteoporosis (loss of bone density). Also called soya and soybean. Latin name: *Glycine max*.

statin

A drug used to treat high cholesterol. Statins lower the amount of cholesterol and certain fats in the blood.

stroke

A loss of blood flow to part of the brain. Strokes are caused by blood clots or broken blood vessels in the brain, and result in damage to a section of brain tissue. Symptoms include dizziness, numbness, weakness on one side of the body, and problems with talking or understanding language. The chance (risk) of stroke is increased by high blood pressure, older age, smoking, diabetes, high cholesterol, heart disease, a family history of stroke, and a build-up of fatty material inside the coronary arteries (atherosclerosis). See also NIH publication: Know Stroke. Know the Signs. Act in Time.

<http://www.ninds.nih.gov/disorders/stroke/knowstroke.htm>

supplement

A nutrient that may be added to the diet to increase the intake of that nutrient. Sometimes used to mean dietary supplement.

synthetic

Made by combining parts to make a whole; usually having to do with substances that are artificial or manufactured.

ultraviolet light

Invisible rays that are part of the energy that comes from the sun. Ultraviolet light also comes from sun lamps and tanning beds. It can damage the skin and cause skin cancer.

upper limit

UL. The largest daily intake of a nutrient considered safe for most people. Taking more than the UL is not recommended and may be harmful. The UL for each nutrient is set by the Food and Nutrition Board at the National Academies of Sciences, Engineering, and Medicine. For example, the UL for vitamin A is 3,000 micrograms/day. Women who consume more than this amount every day shortly before or during pregnancy have an increased chance (risk) of having a baby with a birth defect. Also called the tolerable upper intake level.

US Department of Agriculture

USDA promotes America's health through food and nutrition, and advances the science of nutrition by monitoring food and nutrient consumption and updating nutrient requirements and food composition data. USDA is responsible for food safety, improving nutrition and health by providing food assistance and nutrition education, expanding markets for agricultural products, managing and protecting US public and private lands, and providing financial programs to improve the economy and quality of rural American life.

virus

An organism that can grow and multiply only inside the cells of living humans, plants, or animals. It is able to change (mutate) as it multiplies, which makes viral illnesses difficult to treat. Viruses cause many infections and diseases such as the common cold, AIDS (acquired immunodeficiency syndrome), herpes, and hepatitis.

vitamin

A nutrient that the body needs in small amounts to function and maintain health. Examples are vitamins A, C, and E.

vitamin C

A nutrient needed by the body to make collagen (a protein found in cartilage, tendons, ligaments, bone, and blood vessels), to absorb iron from food, and for wound healing. It is an antioxidant and protects cells from free radical damage. Vitamin C is found in some foods including citrus fruits, strawberries, peppers, dark green vegetables, tomatoes, and potatoes. Also called ascorbic acid.

vitamin E

A nutrient needed by the body to help keep the immune system healthy and to repair damage to DNA. It is an antioxidant that protects cells from free radical damage. Vitamin E is found in some foods, including vegetable oils, nuts and seeds, fortified breakfast cereals, and spinach, broccoli, kiwi, and mangos.

zinc

A mineral found in most cells of the body. It helps enzymes work properly, helps maintain a healthy immune system, helps maintain the senses of taste and smell, and is needed for wound healing, making DNA, and normal growth and development during pregnancy, childhood, and adolescence. Zinc is found in some foods, including oysters, red meat, poultry, beans, nuts, certain seafood, whole grains, fortified breakfast cereals, and dairy products.

Updated: March 22, 2021